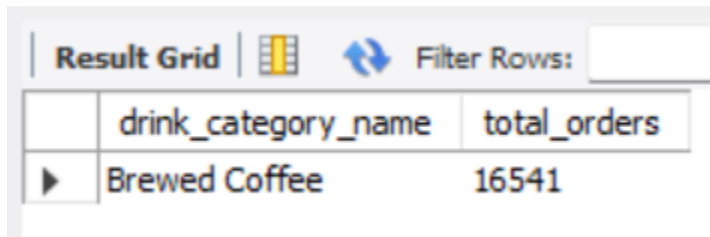


Basic SQL Queries

1) Which drink is the least popular among the customers?

It will join cart_details with drink_category_lookup, then counts how many times each drink category appears, sorts from lowest to highest and picks the least ordered drink

```
SELECT
    d.drink_category_name,
    COUNT(*) AS total_orders
FROM cart_details c
JOIN drink_category_lookup d
    ON c.drink_category_id = d.drink_category_id
GROUP BY d.drink_category_name
ORDER BY total_orders ASC
LIMIT 1;
```



The screenshot shows a database interface with a 'Result Grid' tab. It contains a single row of data representing the least popular drink. The columns are 'drink_category_name' and 'total_orders'. The data row shows 'Brewed Coffee' with 16541 orders.

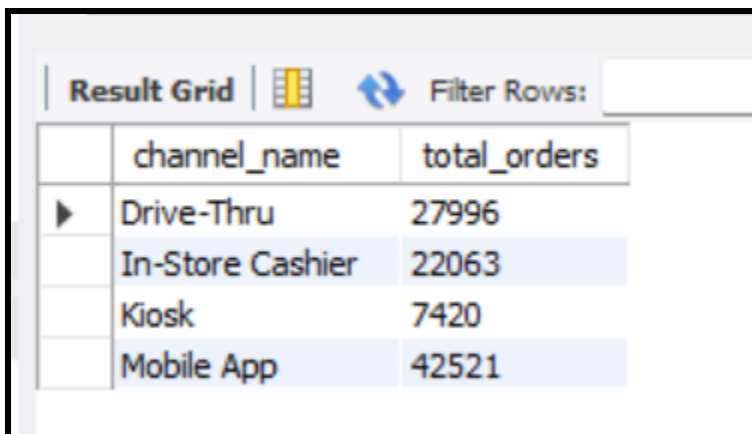
	drink_category_name	total_orders
▶	Brewed Coffee	16541

2) *Count total order by Channel*

This tells you how many orders came from:

- Drive-Thru
- Mobile App
- Kiosk
- In-Store Cashier

```
SELECT
    c.channel_name,
    COUNT(o.order_id) AS total_orders
FROM orders o
JOIN channel_lookup c
    ON o.channel_id = c.channel_id
GROUP BY c.channel_name;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query, showing the channel name and the total number of orders for each channel. The data is as follows:

	channel_name	total_orders
▶	Drive-Thru	27996
	In-Store Cashier	22063
	Kiosk	7420
	Mobile App	42521

3) *Top 10 customers with highest cart size*

This query shows the top 10 orders with highest cart size. A customer may appear multiple times if they placed multiple large orders.

```
SELECT
    cu.customer_id,
    cu.customer_gender,
    cu.customer_age_group,
    c.cart_size,
    o.order_id
FROM cart_details c
JOIN orders o
    ON c.order_id = o.order_id
JOIN customer cu
    ON o.customer_id = cu.customer_id
ORDER BY c.cart_size DESC
LIMIT 10;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 					
	customer_id	customer_gender	customer_age_group	cart_size	order_id
▶	CUST_09068	Female	25-34	10	ORD_00097168
	CUST_10051	Female	25-34	10	ORD_00097414
	CUST_08420	Female	45-54	10	ORD_00097776
	CUST_00098	Female	25-34	10	ORD_00097852
	CUST_14392	Non-binary	45-54	10	ORD_00098024
	CUST_04367	Male	45-54	10	ORD_00098149
	CUST_08266	Male	45-54	10	ORD_00098372
	CUST_11193	Male	18-24	10	ORD_00098513
	CUST_07948	Female	25-34	10	ORD_00099174
	CUST_00741	Male	35-44	10	ORD_00099536

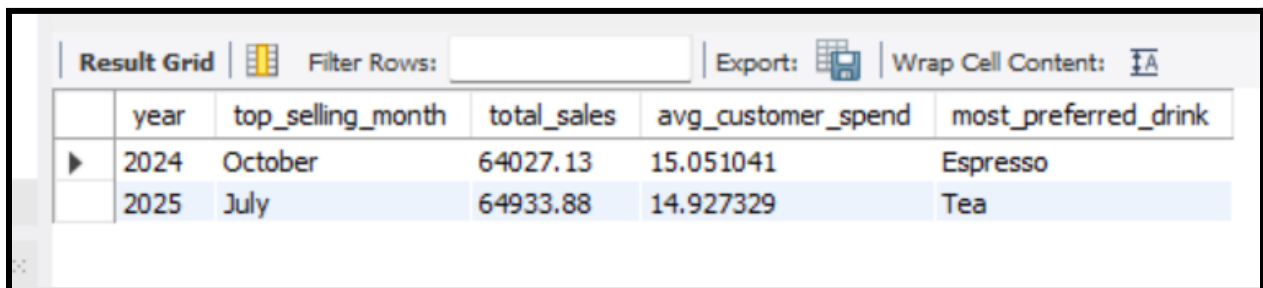
Advance Query

Identify the years present in the dataset, find the top-selling month within each year, calculate the average customer spend for that month, and determine the most preferred drink category.

Answer:

Explanation

This query helps analyze sales trends year by year and shows not only the best-performing month, but also customer spending behavior and drink preference during that month. It gives a meaningful business insight from the Starbucks customer ordering dataset.



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a query. The columns are: year, top_selling_month, total_sales, avg_customer_spend, and most_preferred_drink. The data is as follows:

	year	top_selling_month	total_sales	avg_customer_spend	most_preferred_drink
▶	2024	October	64027.13	15.051041	Espresso
	2025	July	64933.88	14.927329	Tea

I used three tables here: orders, cart_details and drink_category_lookup

Because

Order has information like order_date and total_spend

Cart_details connects the order table by order_id as foreign key and has drink_category_id

Drink_category_lookup converts the drink category ID into the actual drink category name.

This advanced query used the following SQL concepts:

SQL concepts used here:

CTE (WITH clause)

GROUP BY

aggregate functions: SUM(), AVG(), COUNT()

date functions: YEAR(), MONTH(), MONTHNAME()

window function: RANK(), JOIN

```

WITH monthly_sales AS (
    SELECT
        YEAR(o.order_date) AS sales_year,
        MONTH(o.order_date) AS sales_month,
        MONTHNAME(o.order_date) AS month_name,
        SUM(o.total_spend) AS total_sales,
        AVG(o.total_spend) AS avg_customer_spend
    FROM orders o
    GROUP BY
        YEAR(o.order_date),
        MONTH(o.order_date),
        MONTHNAME(o.order_date)
),
ranked_months AS (
    SELECT
        sales_year,
        sales_month,
        month_name,
        total_sales,
        avg_customer_spend,
        RANK() OVER (
            PARTITION BY sales_year
            ORDER BY total_sales DESC
        ) AS month_rank
    FROM monthly_sales
),
top_month_per_year AS (
    SELECT
        sales_year,
        sales_month,
        month_name,
        total_sales,
        avg_customer_spend
    FROM ranked_months
    WHERE month_rank = 1
),
drink_counts AS (
    SELECT
        YEAR(o.order_date) AS sales_year,
        MONTH(o.order_date) AS sales_month,
        d.drink_category_name,
        COUNT(*) AS total_orders,
        RANK() OVER (
            PARTITION BY YEAR(o.order_date), MONTH(o.order_date)

```

```

        ORDER BY COUNT(*) DESC
    ) AS drink_rank
FROM orders o
JOIN cart_details c
    ON o.order_id = c.order_id
JOIN drink_category_lookup d
    ON c.drink_category_id = d.drink_category_id
GROUP BY
    YEAR(o.order_date),
    MONTH(o.order_date),
    d.drink_category_name
),
top_drink_per_month AS (
    SELECT
        sales_year,
        sales_month,
        drink_category_name,
        total_orders
    FROM drink_counts
    WHERE drink_rank = 1
)
SELECT
    t.sales_year AS year,
    t.month_name AS top_selling_month,
    t.total_sales,
    t.avg_customer_spend,
    d.drink_category_name AS most_preferred_drink
FROM top_month_per_year t
JOIN top_drink_per_month d
    ON t.sales_year = d.sales_year
    AND t.sales_month = d.sales_month
ORDER BY t.sales_year;

```